Transfer Learning With Nonlinear Spectral Synthesis for Hyperspectral Target Detection

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***Abstract*— Spectral distortion severely limits detection performance in hyperspectral imagery, while feature learning with neural networks could provide sufficient capacity to enhance spectral consistency. This article designs an end-to-end hyperspectral target detection (HTD) network based on transfer learning and nonlinear spectral synthesis (TLNSS). We first utilize bilinear mixture model (BMM) to synthesize nonlinear target and background spectra for training sample augmentation, which could better characterize ground objects in complex environments. Due to the mutual constraints between the quantity and diversity of the synthesized spectra, transfer learning is introduced to further address data insufficiency. Specifically, we propose an asymmetric autoencoder with a particularly designed multilevel loss to maximally distinguish the reconstruction residuals of background and target, where the multiscale feature extraction subnetwork is trained with abundant reference data, and the simple restoration subnetwork is updated with the simulated spectra. To effectively reconstruct the input as expected, the features extracted from different blocks are complementarily integrated through residual attention. Finally, we accumulate reconstruction residuals across all levels for final detection. The experimental results and ablation analysis of single-data detection on three hyperspectral images verify the superiority and effectiveness of the proposed method and further cross-data detection consolidates the satisfactory tolerance of TLNSS to spectral variation.**

***Index Terms*— Hyperspectral target detection (HTD), nonlinear spectral synthesis, reconstruction residual, transfer learning.**

I. Introduction

HYPERSPECTRAL remote sensing with both spatial and spectral characteristics could provide more information for distinguishing desired targets from irrelevant backgrounds, which is beneficial to practical applications in marine surveillance [1], [2], [3], search and rescue [4], [5], urban planning [6], [7], [8], and more [9], [10]. However, affected by atmospheric conditions, surface compositions, sensor noise, or other undetermined factors, spectral distortion occurs unavoidably during spectral imaging. This phenomenon brings enormous performance limitations to accurate spectral data interpretation and analysis. As a popular research direction in hyperspectral image processing, target detection aspires to pick out those pixels that have the same spectral characterization as the target prior from complex scenes. In comparison with unsupervised anomaly detection, target detection is well aware of the specific class of the extracted objects, namely, performing in an active manner.

In order to mitigate the negative impacts of spectral distortion on hyperspectral target detection (HTD), many algorithms have been presented. With multiple target priors available, Ren et al. [11] developed the popular constrained energy minimization (CEM) into multiple-target CEM (MTCEM), sum CEM (SCEM), and winner-take-all CEM (WTACEM), which have higher tolerance to spectral variation. Recently, Chang [12] provided an in-depth explanation of CEM from the perspective of signal-to-noise ratio (SNR) theory and also conducted a comprehensive analysis of several classical HTD algorithms with hypothesis testing, SNR, and spectral angle theories. Under similar prior conditions, representation-based studies achieve good performance in dealing with spectral inconsistency, including combined sparse and collaborative representation (CSCR) [13], target dictionary construction-based sparse detector (TDCbSD) [14], and single-spectrum-driven binary-class sparse target detector (SSBSTD) [15]. An iterative strategy was also applied to purify target/background spectrum from limited known spectra, such as reweighted adaptive coherence/cosine estimator (rACE) [16], iterative generalized likelihood ratio test (iGLRT) [17], and hierarchical CEM (hCEM) [18]. Moreover, spatial and spectral information were integrated to strengthen the characteristics of different objects, such as diverse-direction CEM (D^2CEM) [19] and sparse-spatialCEM [20]. Recently, deep learning technology has shown great potential in informative feature extraction and discriminative capacity improvement, which brings significant positive feedback to many tasks, such as classification [21], [22], quality enhancement [23], [24], and detection [25]. Existing deep learning-based HTD methods can be generally split into two categories: single- and two-step methods. The single-step detection performs feature

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extraction and target detection in a unified architecture, while the two-step detection first designs an intrinsic representation learner for feature enhancement and then constructs a separate detector to identify targets of interest. The detailed research progress of deep learning-based HTD is reviewed as follows.

A. Single-Step Detection

In [26], a convolutional neural network-based detection (CNND) was designed to learn similarity–dissimilarity measurement by training *pixel-pair spectral difference* samples constructed from a reference hyperspectral dataset, providing guidance for target and background distinction in testing images. Instead of transferring knowledge from reference data to testing data, Du and Li [27] directly constructed *pixel-pair spectral difference* with prior-known target and background instances for deep model optimization, and achieved outstanding detection performance due to no need for data distribution alignment. Zhang et al. [28] proposed a novel HTD network (HTD-Net) as an improved version of CNND, which introduced a feasible solution to the issue of few training samples. Specifically, it amplifies the number of targets and backgrounds with symmetric autoencoder and linear prediction strategy, respectively. The aforementioned three single-step methods achieve target detection through *pixel-pair spectral difference* learning but all ignore the original spectral information. To address this issue, Zhu et al. [29] presented a two-stream convolutional network for target detection (TSCNTD), which first applies two CNNs to extract hidden features of the target prior and target/background training samples, and then constructed *pixel-pair feature difference* for the last detection decision layer. Gao et al. [30] proposed an auxiliary generative adversarial network (AGAN) for HTD, where the generator simulates target and background spectra for data augmentation, and the discriminator integrates true/false discrimination and target/background classification so that it could directly output the detection probability. Recently, inspired by two parallel branches of the novel Siamese network, several works have attempted to employ it for HTD. Rao et al. [31] proposed a Siamese transformer-based target detection (STTD) network, which outputs the final detection score from *pixel-pair feature difference* in a manner similar to TSCNTD. In addition, STTD generates large amounts of training samples by *linear mixture* between the target prior and high-purity backgrounds extracted through unmixing and refiltering. Zhang et al. [32] ensembled several Siamese fully connected networks for stable target detection (denoted as SFCTD), and it also utilized *linear mixture model* (LMM) for pseudo spectra generation. Different from STTD, SFCTD directly regards each test pixel as background by assuming that the target pixels in the testing scene are far fewer than the complex backgrounds. Recently, a novel deep CEM (DCEM) network was presented, and it consists of several simple fully connected layers for automatic nonlinear filter learning, where the filter satisfies two conditions: 1) the response to the target prior is equal to 1 and 2) the average energy of all testing pixels is minimal. In summary, most existing single-step detection methods generally proceed from three aspects: labeled data generation, robust knowledge transfer, and novel architecture design.

B. Two-Step Detection

Xie et al. et al. [33] addressed the target detection problem from the perspective of background modeling and designed a background learning network based on a target suppression constraint (BLTSC). The proposed BLTSC only trained on background samples could reconstruct backgrounds with small errors, making it possible to locate targets based on the discrepancy between the reconstructed data and the original data. Moreover, Xie et al. [34], [35] employed generative adversarial network (GAN) or variational autoencoder (VAE) to extract useful hidden features from input hyperspectral images and then implemented target detection in the high-level feature space. Zhu et al. [36] proposed a deep metric learning model to explore discriminative interclass features for HTD, and they also presented a new superpixel segmentation-based background generation method and an LMM-based target augmentation method to address the issue of insufficient training data. In [37], a meta-learning-based detector with Siamese network (MLSN) was designed to boost the intraclass spectral similarity and interclass spectral dissimilarity, and the spectral angle between the target prior and each testing pixel in the feature space was taken as the final detection score. Shi et al. [38], [39], [40] designed several novel feature extraction networks based on the outstanding reconstruction performance of autoencoder and strong discriminative capacity of distinctive constraint and then applied a simple matching strategy to the informative features for target identification. Besides the unsupervised autoencoders, Shi et al. [41] also integrated semisupervised domain adaption and few-shot learning for HTD, in which the pairwise similarity measurement learned from the source domain was adaptively transferred to the target domain. Recently, Li et al. [42] proposed a novel hierarchical denoising autoencoder to enhance spectral consistency without dimensionality reduction, which could facilitate detection requiring spectral information at specific wavelengths. In common, the aforementioned works project the original hyperspectral image into a new feature space with greater disparity, enabling easier identification of desired targets from complex backgrounds. Compared with the end-to-end single-step detection, the feature extraction process in circuitous two-step detection is not always foolproof, i.e., there may be some information really important for feature extraction but not very effective for target detection.

This article is oriented toward the end-to-end single-step HTD task, where only several target spectra are given and background is generally required to be extracted from the testing scene itself. Therefore, it is vital to filter out representative background spectra and synthesize more realistic target and background samples to support network training. However, existing works apply pure spectra or linearly mixing spectra, and they fail to accurately describe real scenarios that turn out to be nonlinear interactions [43], [44]. On the other hand, the spectral synthesis using several known target and

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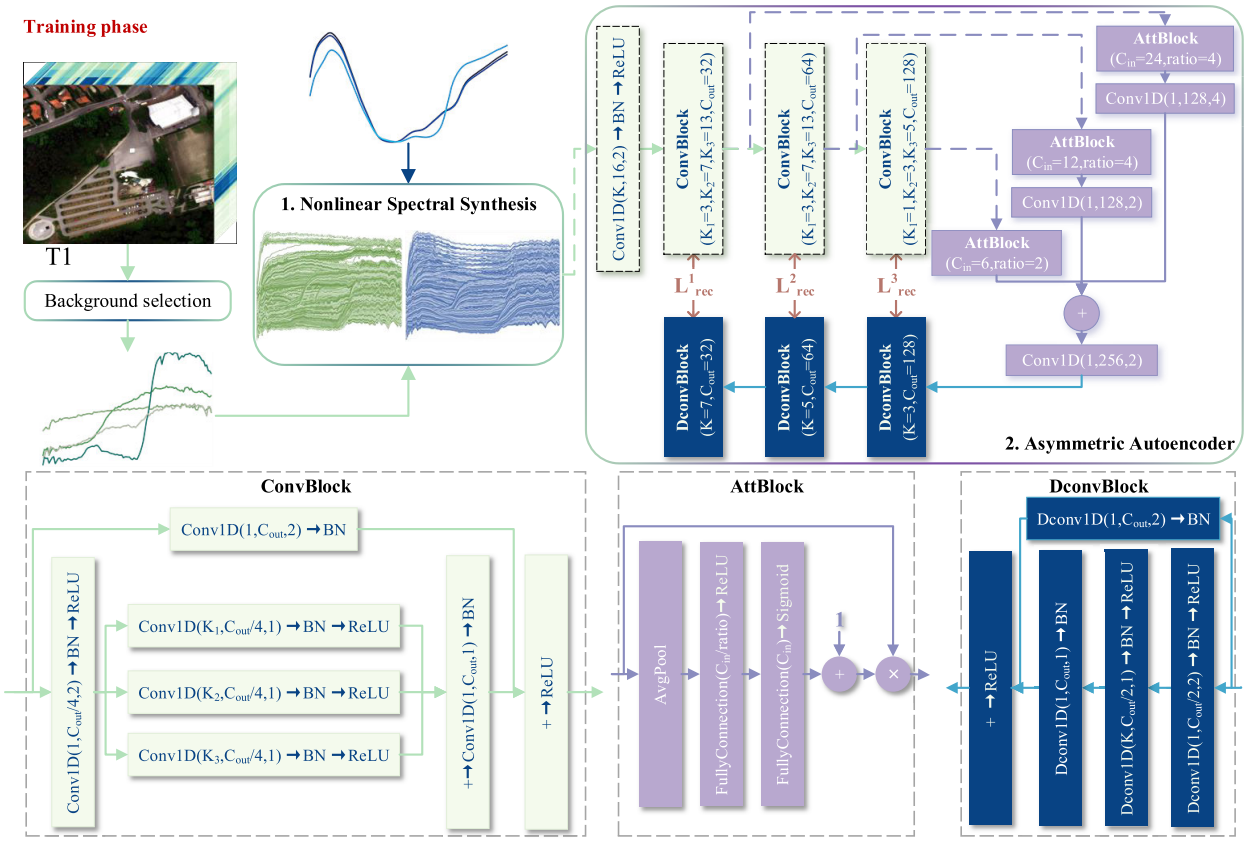


Fig. 1. Overall flowchart of the proposed TLNSS during the training phase, including two parts: nonlinear spectral synthesis for training data generation and asymmetric autoencoder for detection model optimization.

background spectra can only generate samples with limited diversity, which cannot provide enough information for network optimization. To address these issues, we propose an end-to-end transfer learning-based HTD method [denoted as transfer learning and nonlinear spectral synthesis (TLNSS)], which first extracts discriminative features with knowledge from a known domain and further optimizes the detection model by training on nonlinearly synthesized samples. The basic structure of TLNSS is an asymmetric autoencoder, where the discriminative feature extraction subnetwork consists of several multiscale convolution blocks, the multilayer feature fusion subnetwork mainly includes several residual attention blocks, and the reverse spectra restoration subnetwork consists of several simple deconvolution blocks. The feature extraction subnetwork followed by a classifier is trained on sufficient reference data, and the feature fusion and spectra restoration subnetworks are trained on the simulated nonlinear spectra. In order to achieve effective detection, we design a multilevel loss function, which minimizes the reconstruction residuals of backgrounds but maximizes that of targets. The main contributions can be summarized as follows.

1) To address the issue of insufficient training samples, we propose a nonlinear spectral synthesis method to simulate nonlinearly mixing target and background spectra, which is more in line with the real environment. Moreover, transfer learning is adopted to alleviate the interplay between the quantity and diversity of the synthesized spectra.
2) For TLNSS training acceleration, we optimize the feature extraction subnetwork with a complex architecture on the reference data only once and then freeze it to train the remaining succinct feature fusion and spectra restoration subnetworks with the simulated samples. This design contributes to achieving significant efficiency gains.
3) To keep the objectives of network optimization and target detection consistent, the detector is formulated the same as the multilevel loss, which maximizes the reconstruction residual differences between target and background. Furthermore, residual attention is designed to automatically select features favorable for the target task.

The remainder of this article is organized as follows. Section II introduces the proposed TLNSS method in detail. In Section III, experiments and further analysis of single-data detection and cross-data detection are given. Finally, Section IV concludes the research work presented in this article.

II. Proposed Method

Fig. 1 shows the overall flowchart of TLNSS during the training phase. The proposed TLNSS algorithm consists of two main parts: nonlinear spectral synthesis and asymmetric autoencoder. Given the testing hyperspectral image and target priors, we first apply the automatic target generation

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process (ATGP) [45] for representative background selection and then synthesize diverse mixing target and background spectra based on the popular bilinear mixture model (BMM). The asymmetric autoncoder includes multiscale feature extraction, multilayer feature fusion, and reverse spectra restoration, in which the extraction subnetwork is pretrained using the reference data with pixel-level labels for informative feature learning, and the remaining feature fusion and spectra restoration subnetworks are updated with the nonlinearly synthesized data to accumulate multiple feature residuals for final detection. More details are presented in the following.

A. Nonlinear Spectral Synthesis

Achieving HTD using only the target spectra is feasible, but it would bring effective improvements if information about the complicated backgrounds is provided. In practice, pre-knowledge about the backgrounds covering the entire testing scene is often not available, and there are generally two ways to collect reliable background spectra. The first is manual selection from the spectral library (e.g., USGS [46] and KLUM [47]), which contains very fine spectra of many common materials. However, due to different acquisition conditions, spectral misalignment between the given backgrounds and testing images may seriously affect target detection, resulting in unfair performance evaluation. The second is immediate extraction from the testing hyperspectral image, frequently used in existing studies. In this article, ATGP is exploited for reliable background overextraction to characterize the hyperspectral scene as comprehensively as possible.

Let $\mathbf{S}_I$ be a hyperspectral image with $N_h \times N_w$ pixels and N_l bands, and $\mathbf{S}_T \in \mathbb{R}^{N_l \times N_t}$ be N_t known target spectra. ATGP searches N_b backgrounds $\mathbf{S}_B$ from $\mathbf{S}_I$ with high discrepancy to $\mathbf{S}_T$ through orthogonal subspace, formulated as

$$\mathbf{P}_{\mathbf{U}}^{\perp} = \mathbf{I} - \mathbf{U}\left(\mathbf{U}^T\mathbf{U}\right)^{-1}\mathbf{U}^T \tag{1}$$

where $\mathbf{I}$ is the identity matrix and $\mathbf{P}_{\mathbf{U}}^{\perp}$ is orthogonal to $\mathbf{U}$, i.e., $\mathbf{P}_{\mathbf{U}}^{\perp}\mathbf{U} = \mathbf{0}$.

The number of the overly extracted backgrounds denoted as N_b is set to βn_b, where n_b represents the number of potential backgrounds estimated with the well-known Harsanyi–Farrand–Chang method [48], and β is a scale factor no less than 1. Section III-A2 further analyzes the impacts of β on detection performance. With the aforementioned definitions, ATGP proceeds in an iterative way until the stop condition is triggered, as shown in Algorithm 1.

Training detection model with the available targets $\mathbf{S}_T$ and backgrounds $\mathbf{S}_B$ might lead to obvious overfitting, so it is necessary to simulate more samples. Considering that interactions among multiple materials are quite common during hyperspectral imaging, as well as data characterization by LMM is not very accurate in this case, we apply BMM to generate mixing spectra under different abundance fractions. Given two spectral vectors $\mathbf{s}^1$ and $\mathbf{s}^2$, BMM is formulated as

$$\mathbf{s} = \underbrace{\alpha\mathbf{s}^1 + (1-\alpha)\mathbf{s}^2}_{\text{linear}} + \underbrace{\alpha(1-\alpha)\mathbf{s}^1 \odot \mathbf{s}^2}_{\text{bilinear}} \tag{2}$$

Algorithm 1 ATGP Algorithm

Require: Hyperspectral image $\mathbf{S}_I$, prior-known target spectra $\mathbf{S}_T$, and the number of background spectra to be extracted N_b

Ensure: Background spectra $\mathbf{S}_B$

Initialization: iteration $k = 1$, $\mathbf{S}_B^{(k-1)} = [\ ]$, $\mathbf{S}_U^{(k-1)} = \mathbf{S}_T$.
while $k < N_b$ **do**
  $\mathbf{P}_{\mathbf{S}_U^{(k-1)}}^{\perp} = \mathbf{I} - \mathbf{S}_U^{(k-1)}\left(\mathbf{S}_U^{(k-1)^T}\mathbf{S}_U^{(k-1)}\right)^{-1}\mathbf{S}_U^{(k-1)^T}$
  $\mathbf{s}_B^{(k)} = \arg\max_{\mathbf{s}_I^j}\left(\mathbf{P}_{\mathbf{S}_U^{(k-1)}}^{\perp}\mathbf{s}_I^j\right)^T\left(\mathbf{P}_{\mathbf{S}_U^{(k-1)}}^{\perp}\mathbf{s}_I^j\right)$
  $\mathbf{S}_U^{(k)} \leftarrow \left[\mathbf{s}_U^{(k-1)}\ \mathbf{s}_B^{(k)}\right]$
  $\mathbf{S}_B^{(k)} \leftarrow [\mathbf{s}_B^{(k-1)}\ \mathbf{s}_B^{(k)}]$
  $k \leftarrow k + 1$
end while
return $\mathbf{S}_B^{(k-1)}$

where α denotes the fraction value. For target synthesis, α is selected from $\{0.05, 0.3\}$ with an interval of δ_1, and $\mathbf{s}^1$ and $\mathbf{s}^2$ represent the spectra of background and target, respectively. For background synthesis, α is selected from $\{0.1, 1.0\}$ with an interval of δ_2, and both $\mathbf{s}^1$ and $\mathbf{s}^2$ are backgrounds. Note that δ_1 and δ_2 are set in coordination with N_t and N_b to keep the number of target and background spectra roughly equal.

B. Asymmetric Autoencoder

The asymmetric autoencoder aims to extract shallow-layer details and deep-layer abstract features from the training spectra and then restore different layerwise features from the multilayer fused feature. As shown in Fig. 1, the feature extraction subnetwork with three ConvBlocks is more complicated than the spectra restoration subnetwork with three DconvBlocks, which could learn multiscale representative information and facilitate feature restoration with high efficiency. Considering that the number of simulated samples is probably insufficient to support robust network optimization, we train the feature extraction subnetwork using a reference classification dataset with abundant labels and then apply the frozen weights to explore discriminative features for detection based on transfer learning. The remaining feature fusion and spectra restoration subnetworks with less parameters could be well optimized using the simulated nonlinear spectra.

1) Multiscale Feature Extraction Subnetwork: The feature extraction subnetwork is marked in light green in Fig. 1, including one convolution layer and three ConvBlocks. The first convolution layer is designed for noise removal and feature enhancement through automatic weighting. Then, three ConvBlocks with residual learning and multiscale kernels are applied to extract intrinsic spectral features, which could exactly enhance distinctions between different materials. In specific, given an input feature $\mathbf{F} \in \mathbb{R}^{n_c \times n_l}$ with n_c channels and n_l bands, each ConvBlock performs as follows:

$$\mathbf{F}_e^1 = \mathbf{F} *_{br} \boldsymbol{\omega}_{(1,n_c^o/4,2)} \tag{3a}$$

$$\mathbf{F}_e^m = \left(\sum_{i=1}^{3} \mathbf{F}_e^1 *_{br} \boldsymbol{\omega}_{(k_i,n_c^o/4,1)}\right) *_b \boldsymbol{\omega}_{(1,n_c^o,1)} \tag{3b}$$

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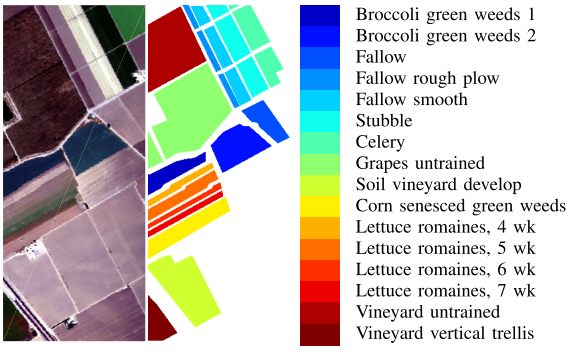


Fig. 2. Visualization of Salinas data. From left to right: pseudo-color image, ground-truth map, and legends.

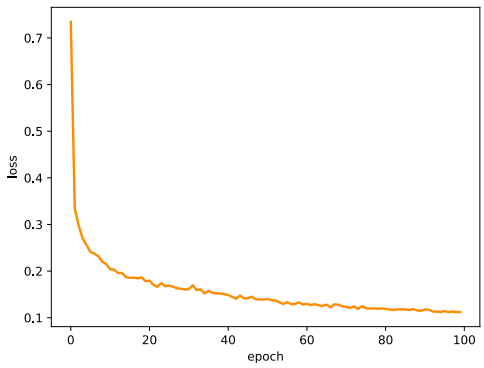


Fig. 3. Training loss curve of the multiscale feature extraction-classifier network on the Salinas dataset.

$$\mathbf{F}_e^s = \mathbf{F} *_b \boldsymbol{\omega}_{(1,n_c^o,2)} \tag{3c}$$

$$\mathbf{F}_e^o = f_{\text{relu}}(\mathbf{F}_e^s + \mathbf{F}_e^m) \tag{3d}$$

where the convolution kernel is denoted as $\boldsymbol{\omega}_{(k,n_c^o,s)}$; k, n_c^o, and s are kernel size, output channel, and stride, respectively; $\mathbf{F}_e^m$, $\mathbf{F}_e^s$, and $\mathbf{F}_e^o$ are output features of the main branch, shortcut branch, and ConvBlock, respectively; $f_{\text{relu}}(\cdot)$ represents nonlinear activate function; $*_b$ is the 1-D convolution followed by batch normalization; and $*_{br}$ refers to $*_b$ followed by $f_{\text{relu}}(\cdot)$.

As mentioned before, in order to obtain a well-trained feature extraction network, we introduce a simple classifier consisting of two fully connected layers after the third ConvBlock to construct a supervised classification network and then minimize the cross-entropy loss between the ideal label and the predicted score. In this article, a popular hyperspectral dataset Salinas acquired by Airborne Visible/InfraRed Imaging Spectrometer (AVIRIS) over Salinas Valley, CA, USA, is applied for network training, as visualized in Fig. 2. The Salinas data include several land-cover classes with similar spectral reflectance, which helps generate a powerful feature extractor. Fig. 3 shows the loss curve during network training on Salinas. We initialize the learning rate as 0.005 and change its value as training progresses. It can be observed that the cross-entropy loss initially decreases significantly and then converges to a small value at a slow but continuously decreasing speed. When obtaining the trained network, the classifier would be removed, and the former feature extraction subnetwork with fixed parameters is concatenated with the following feature fusion and spectra restoration subnetworks for target detection.

2) Multilayer Feature Fusion Subnetwork: It is possible to directly restore multilayer representations from the output of feature extraction subnetwork. However, the learned deep features mostly reflect high-level semantic information, and it would bring challenges for low-level feature recovery. To address this issue, we present a multilayer feature fusion subnetwork consisting of several AttBlocks and convolution operations, which could capture complementary features from different layers through adaptive accumulation, as marked in light purple in Fig. 1. To overcome the loss of useful features caused by occasionally inappropriate weights, the introduced AttBlock is equipped with a robust residual structure. Specifically, Attblock first averages the features along the channel with AvgPool and generates weight values through two fully connected layers followed by the ReLU or Sigmoid activator. Then, the elementwise sum and product operations are applied to output representative features. Moreover, in order to fuse multilayer features with different sizes, we downsample the low-level features along the spectral dimension and upsample them along the channel dimension through a convolution layer. Let $\mathbf{F}_e^{o^i} \in \mathbb{R}^{n_c^i \times n_l^i}$ be the output of the ith ConvBlock, and the fusion process can be denoted as follows:

$$\mathbf{F}_f = \left[\sum_{i=1}^{2} f_{Att}(\mathbf{F}_e^{o^i}, n_c^i, r_i) * \boldsymbol{\omega}_{(1,128,s_i)} + f_{Att}(\mathbf{F}_e^{o^3})\right] *\boldsymbol{\omega}_{(1,256,2)} \tag{4}$$

where r_i and s_i are ratio and stride, respectively; $*$ denotes fully connected operation; and $f_{Att}(\cdot)$ is the function, including AttBlock and following convolution operation for scale transformation. Note that the last convolution layer $\boldsymbol{\omega}_{(1,256,2)}$ is exploited to obtain informative and concentrated features.

3) Reverse Spectra Restoration Subnetwork: As marked in dark blue in Fig. 1, the reverse spectra restoration subnetwork reconstructs representations in correspondence with the extraction subnetwork, which similarly consists of three DconvBlocks to allow pairwise feature residual calculation. Although DconvBlock applies a residual-like structure the same as Convblock, it only captures single-scale features, resulting in a lightweight subnetwork. This asymmetric structure relaxes the requirement for large amounts of training data. For the fused feature $\mathbf{F}_f$, the restoration subnetwork implements in the following fashion:

$$\mathbf{F}_r^m = \mathbf{F}_f \tilde{*}_{br} \boldsymbol{\omega}_{(1,n_c^o/2,2)} \tilde{*}_{br} \boldsymbol{\omega}_{(k,n_c^o/2,1)} \tilde{*}_b \boldsymbol{\omega}_{(1,n_c^o,1)} \tag{5a}$$

$$\mathbf{F}_r^s = \mathbf{F}_f \tilde{*}_b \boldsymbol{\omega}_{(1,n_c^o,2)} \tag{5b}$$

$$\mathbf{F}_r^o = f_{\text{relu}}(\mathbf{F}_r^s + \mathbf{F}_r^m) \tag{5c}$$

where $\mathbf{F}_r^m$, $\mathbf{F}_r^s$, and $\mathbf{F}_r^o$ are the output features corresponding to the main branch, shortcut branch, and DconvBlock, respectively; $\tilde{*}_b$ represent the 1-D deconvolution followed by batch normalization; and $\tilde{*}_{br}$ refers to $\tilde{*}_b$ followed by $f_{\text{relu}}(\cdot)$.

For parameter optimization of the feature fusion and restoration subnetworks, we construct a multilevel loss based on spectral angle. Given a training spectrum $\mathbf{s}$ labeled with g ($g = 1$ for target and $g = 0$ for background), let $\mathbf{F}_e^{o^i}$, $\mathbf{F}_r^{o^i} \in \mathbb{R}^{n_c^i \times n_l^i}$ indicate the pairwise features from the ith ConvBlock

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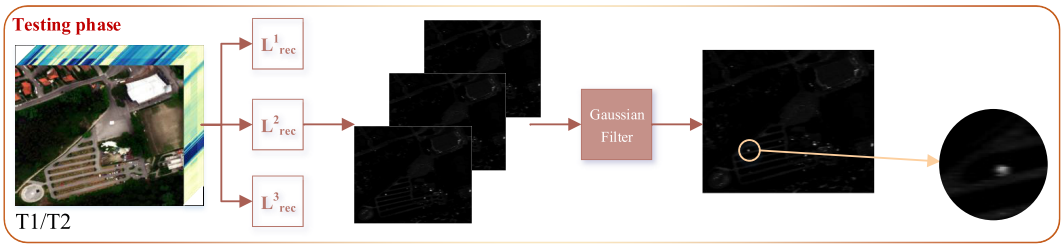


Fig. 4. Overall flowchart of the proposed TLNSS during the testing phase.

and DconvBlock, $i = 1, 2, 3$, and we calculate the cosine similarity along the band axis and obtain the loss value by averaging along the channel dimension. Mathematically

$$\text{loss} = \frac{1}{3}\sum_{i=1}^{3}\left\{\frac{1}{n_c^i}\sum_{j=1}^{n_c^i}\left[(g == 1)?d_j^i : (1 - d_j^i)\right]\right\} \tag{6a}$$

$$d_j^i = \frac{\left(\mathbf{F}_e^{o^i}(j,:)\right)^{\mathrm{T}} \cdot \mathbf{F}_r^{o^i}(j,:)}{\left\|\mathbf{F}_e^{o^i}(j,:)\right\|_2 \left\|\mathbf{F}_r^{o^i}(j,:)\right\|_2} \tag{6b}$$

in which d_j^i denotes the cosine similarity between the jth channel of $\mathbf{F}_e^{o^i}$ and $\mathbf{F}_r^{o^i}$. The expression $\langle e_1\rangle?\langle e_2\rangle:\langle e_3\rangle$ means that if the condition e_1 holds, then execute e_2; otherwise, execute e_3. The proposed TLNSS method commits to minimize the loss function on all training samples. By observing from (6a), the reconstruction residual $1 - d_j^i$ would be maximized for target spectra and minimized for background spectra, i.e., successfully identifying interested targets from irrelevant backgrounds.

C. Target Detection

During the testing phase, we predict the probability that each pixel being a target with cosine measurement the same as (6b), as shown in Fig. 4. For hyperspectral image $\mathbf{S}_I' \in \mathbb{R}^{N_h' \times N_w' \times N_l'}$, the detection score for the kth pixel can be calculated as

$$y_k = \sum_{i=1}^{3}\frac{1}{n_c^i}\sum_{j=1}^{n_c^i}\left[1 - \frac{\left(\mathbf{F}_e^{k^i}(j,:)\right)^{T} \cdot \mathbf{F}_r^{k^i}(j,:)}{\left\|\mathbf{F}_e^{k^i}(j,:)\right\|_2 \left\|\mathbf{F}_r^{k^i}(j,:)\right\|_2}\right]$$
$$k = 1, \ldots, N_h' \times N_w' \tag{7}$$

where $\mathbf{F}_e^{k^i}$ and $\mathbf{F}_r^{k^i}$ are representations of the ith ConvBlock and DconvBlock for the kth pixel, respectively. In order to avoid mistaking noisy pixels as target, we further smooth the detection map $\mathbf{Y}$ with Gaussian filter (GF).

III. Experiments

A. Preparation

1) Datasets: In the experiments, we apply four hyperspectral images, whose characteristics are reported in Table I. Viarregio T1 and T2 were collected by an airborne spectrometer SIM.GA over Viareggio, Tuscany, Italy, at different times [49], and both images have the same spatial and spectral size but different acquisition times. The third image was captured by AVIRIS over San Diego airport, CA, USA. The fourth dataset was preprocessed by pansharpening of a 30-m hyperspectral image and a 5-m panchromatic image acquired by the Italian PRISMA satellite over Pavia, Lombardia, Italy.

TABLE I
Characteristics of the Testing Hyperspectral Images

Images	Time[1]	Date	SpeRes[2]	SpeSiz[2]	SpaRes[2]	SpaSiz[2]
Viareggio T1	14:18	May 8, 2013	0.4-1μm 1.2nm	110	0.6m	375×450
Viareggio T2	14:33					
San Diego	-	-	0.4-2.5μm 10nm	189	3.5m	100×100
Pavia	10:30	April 2, 2021	0.4-2.5μm 10nm	185	5m	744×834

[1] The times are Greenwich Mean Time (GMT).
[2] SpeRes, SpeSiz, SpaRes, and SpaSiz are abbreviations of spectral resolution, spectral size, spatial resolution, and spatial size, respectively.

To support detection performance verification, there are five man-made objects deployed in two Viareggio scenes, namely, NATO Galileo green panel (P1), UNIPI green pressed cardboard panel (P2), blue Ford Fiesta (V1), green Fiat Ducato (V3), and white Ford Focus covered with red tarp (V4). The reflectance spectra of these targets are captured by the FieldSpec handheld spectroradiometer ranging from 0.35 to 2.5 μm with 2151 bands, and 110 bands are preserved to align spectral reflectance with Viareggio data. Fig. 5 shows the spectral curves and photographs of five targets and marks their locations in Viareggio T1 and T2. As for three vehicles, spectra at roof and hood are acquired for comprehensive characteristics description. In addition, there are a total of three P1s, one P2, one V1, one V3, and one V4, and the locations of V1 at T1 and T2 are different. The third dataset contains three airplanes as targets, and Fig. 5 shows the ground-truth map and spectral information sampled from the dataset itself for performance evaluation and target detection. The Pavia data are constructed to identify buildings with asbestos roofs, and the pansharpening preprocessing and acquisition of asbestos spectra are available in our previous work [50]. Fig. 5 also gives the distribution of target pixels and spectral curves of two typical asbestos.

2) Experimental Setup: Evaluation indexes are important to verify the effectiveness of different algorithms. In this article, the grayscale detection map, receiver operating characteristic (ROC) curve, and separability map are utilized for qualitative assessment, in which the ROC curve is specified by false alarm rate P_f and detection probability P_d under continuous thresholds, and the separability map is plotted through

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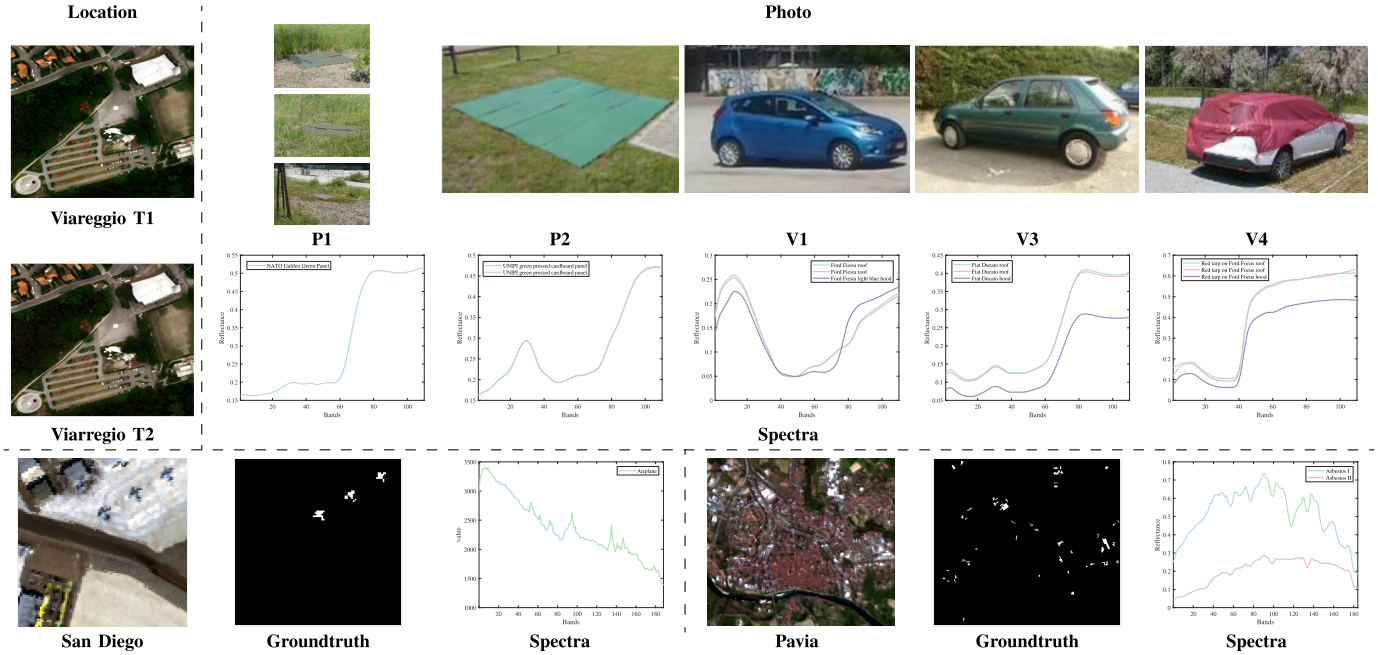


Fig. 5. Locations, photographs, and spectral information of four experimental datasets.

the modified boxplot function. For quantitative evaluation, we apply the popular index, i.e., area under ROC curves (AUC). The larger the AUC, the better the overall detection performance is. Furthermore, as the interested targets in some datasets occupy only several pixels, we also focus on the minimal P_f when all target pixels are correctly identified, thereby characterizing the probability of mistaking background as target with $P_f^{\min}(P_d = 1)$. A good detector should have as small $P_f^{\min}(P_d = 1)$ as possible. Since different algorithms might exhibit varying performance on diverse hyperspectral images, we additionally apply the Friedman test and Nemenyi post hoc test to provide a comprehensive global performance comparison, as in [42].

In order to evaluate the performance of the proposed TLNSS method, several popular algorithms are applied for fair comparison, including classical CEM [51], adaptive coherence/cosine estimator (ACE) [52], recent SNR-based NGDS-SNR and NGR-SNR, and spectral angle-based DS-SA^2 and R-SA^2 [12], machine learning-based CSCR [13], and deep learning-based CNND and HTD-Net. In general, the above comparing methods except CSCR are based on a single spectrum for target detection. Therefore, for targets such as V1 with multiple prior-known spectra, we sum all detection results calculated with each spectrum to obtain a final map. Moreover, in addition to routine performance evaluation on Viareggio T1, San Diego, and Pavia, we implement target identification on Viareggio T2 with detectors constructed using Viareggio T1 to evaluate the effectiveness of TLNSS under harsher spectral distortion conditions.

Parameter setup has significant effects on detection performance. For six statistical detection methods including CEM, ACE, NGDS-SNR, NGR-SNR, DS-SA^2, and R-SA^2, no predetermined parameters are required. For CSCR, the regularization parameters λ_1 and λ_2 are determined by a grid search strategy, and the window size $(w_{\text{in}}, w_{\text{out}})$ is selected according to the target size. For CNND, the learning rate is selected from $\{0.0001, 0.0005, 0.001, 0.003, 0.005, 0.007, 0.01, 0.05\}$, and the best results are displayed for comparison. For HTD-Net, we generate 150 target samples and 100 background samples and construct 11 175 positive pixel-pairs and 15 000 negative pixel-pairs for network training. In our proposed TLNSS method, the training set consists of the prior-known target, extracted background, and synthesized nonlinear samples, while the testing set is the whole hyperspectral image. There are three main parameters: the scale factor β for background generation, the standard deviation σ of GF, and the way to update the learning rate. First, we perform target detection by setting β to 1, 2, 3, and 4, and Fig. 6(a) shows the impacts of β selection on detection performance. In most cases, the accuracy, especially the scaled $P_f^{\min}(P_d = 1)$, improves steadily as β increases until reaching a bottleneck. For convenience, we uniformly set β to 3 for all datasets to characterize the hyperspectral scene as comprehensively as possible. In order to illustrate the effects of GF on detection performance improvement, Fig. 6(b) show the quantitative indexes without GF and with GF at different σ's. It can be found that GF could help to achieve better performance, but its value varies for different hyperspectral images. By analyzing the relationship between the most appropriate σ and the testing datasets, we recommend determining σ according to the target size, usually larger σ for large objects. For the learning rate setup, we initialize it to 0.005 and usually cut it in half or other factors less than 1 if the loss stops falling for five epochs. Fig. 7 plots the loss curves during TLNSS training on Viareggio T1, San Diego, and Pavia. It is common that the loss drops fast at the first several epochs and then tends to slow down as the learning rate decreases, which helps to find the optimal detection model. Moreover, to effectively avoid

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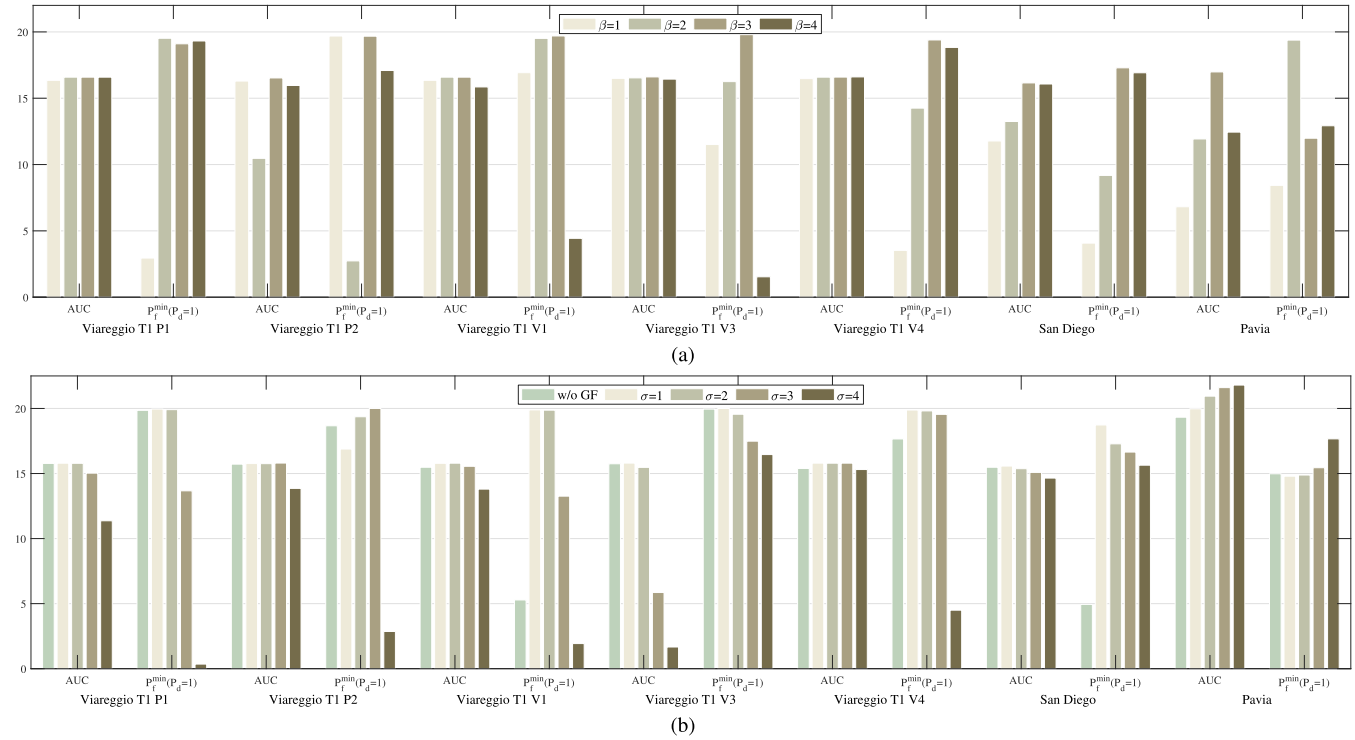


Fig. 6. Impacts of parameter setup on detection performance. For clarity, we scale the accuracy values by different coefficients, and the sign of $P_f^{min}(P_d = 1)$ is further reversed, and thus, the larger the better. (a) Scale factor β for background generation. (b) Standard deviation σ of GF.

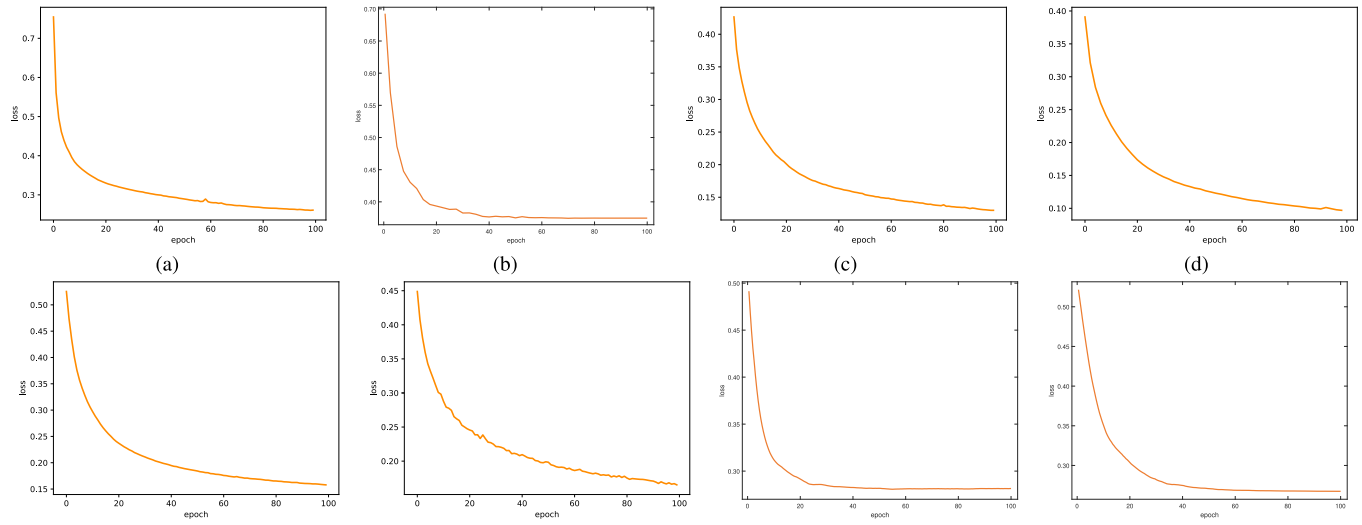


Fig. 7. Training loss curves for multilayer feature fusion and reverse spectra restoration subnetworks on the experimental datasets. Note that only the test loss on P1 is shown here to save space. (a) Viareggio T1 P1. (b) Viareggio T1 P1 (test). (c) Viareggio T1 P2. (d) Viareggio T1 V1. (e) Viareggio T1 V3. (f) Viareggio T1 V4. (g) San Diego. (h) Pavia.

overfitting problem, we monitor the loss values of the testing data during network training, and the results in Fig. 7(b) validate the outstanding generalization performance of TLNSS. It should be emphasized that the testing loss never participates in network optimization. Furthermore, we implement all experiments on a workstation with an NVIDIA GeForce GTX Titan Xp GPU, using Pytorch for CNND and TLNSS, Tensorflow for HTD-Net, and MATLAB for the others.

B. Experimental Results and Discussion

1) Single-Data Detection: Fig. 8 shows the detection maps of different comparing methods on Viareggio T1, San Diego, and Pavia. Taking a global view, the statistical methods and our TLNSS suppress background well, while CSCR, CNND, and HTD-Net show relatively large values for background pixels. By zooming in on the target area, the statistical detectors would omit some target pixels at the edge and also produce some false alarms, which can be found from the detection maps of airplane by ACE, NGDS-SNR, and NGR-SNR, and V4 by DS-SA^2 and R-SA^2. CSCR, CNND, and HTD-Net show unacceptable visualization performance on small objects and slightly improved effects but still unsatisfactory on large targets. TLNSS could successfully detect the edge pixels with GF and give better distinctions between target and background due to the adequate diversity of training samples. However,

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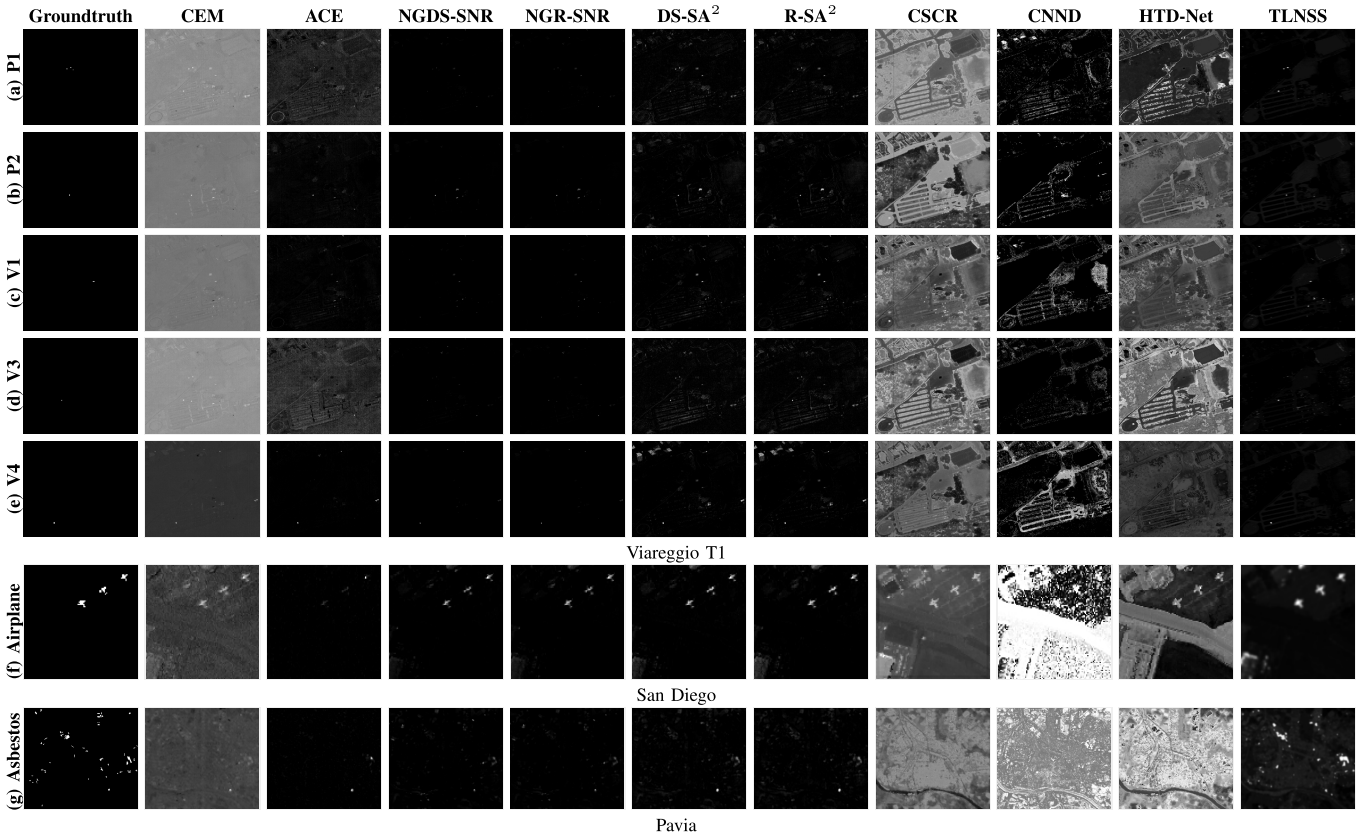


Fig. 8. Detection maps of different competitors on the experimental datasets. From left to right: ground truth, detection maps of CEM, ACE, NGDS-SNR, NGR-SNR, DS-SA^2, R-SA^2, CSCR, CNND, HTD-Net, and TLNSS. Best view via zoomed-in view. (a) P1. (b) P2. (c) V1. (d) V3. (e) V4. (f) Airplane. (g) Asbestos.

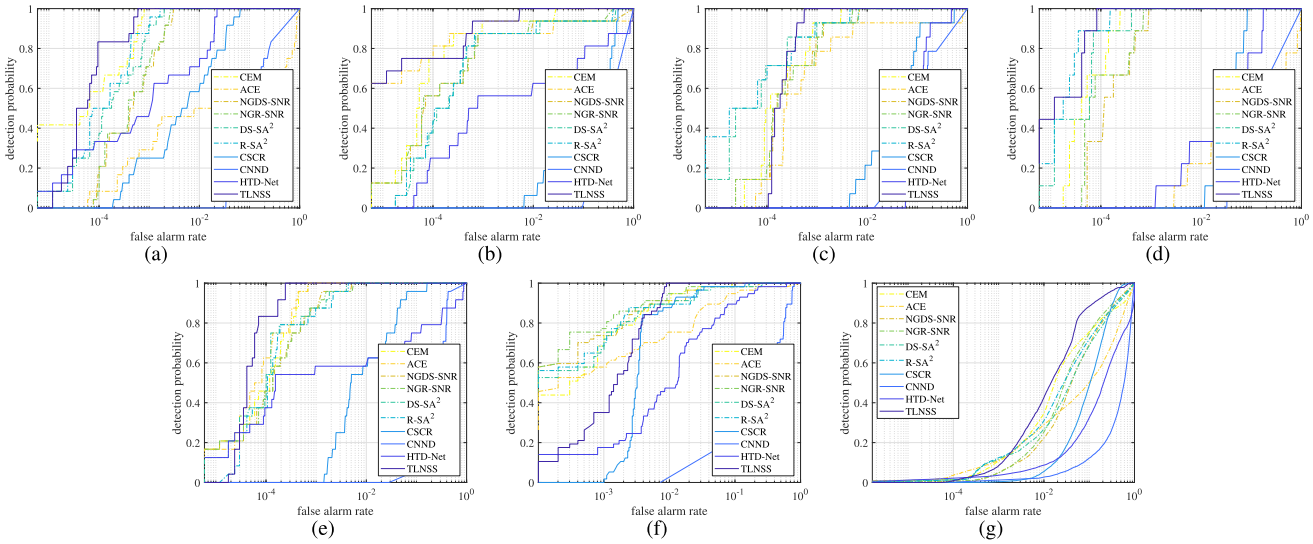


Fig. 9. ROC curves of different detection algorithms on the experimental datasets. (a) Viareggio T1 P1. (b) Viareggio T1 P2. (c) Viareggio T1 V1. (d) Viareggio T1 V3. (e) Viareggio T1 V4. (f) San Diego. (g) Pavia.

as the nonlinear spectral synthesis can only simulate the complex environment as closely as possible and cannot fully cover all cases, there are also more or less false alarms in TLNSS.

To further clarify the relationship between false alarm rate and detection probability, Figs. 9 and 10 plot the ROC curves and separability maps of different detection algorithms on Viareggio T1, San Diego, and Pavia, respectively. Clearly, the ROC curves of TLNSS and the statistical methods except ACE are much closer to the top-left corner of the axis than others in most cases, indicating that they obtain better detection performance. TLNSS occasionally obtains a smaller detection probability at a very low P_f but generally has a higher detection probability when the false alarm rate is

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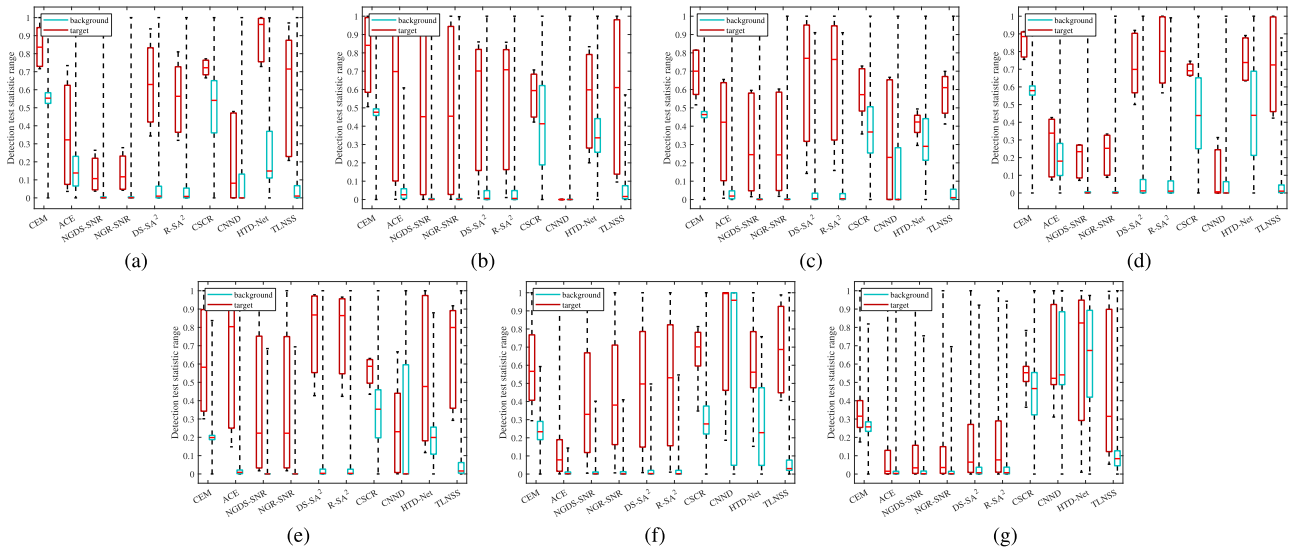


Fig. 10. Separability maps of different detection algorithms on the experimental datasets. (a) Viareggio T1 P1. (b) Viareggio T1 P2. (c) Viareggio T1 V1. (d) Viareggio T1 V3. (e) Viareggio T1 V4. (f) San Diego. (g) Pavia.

greater than 10^{-2}. The separability maps show that there are obvious overlaps between the target boxplot and the background boxplot of CSCR, CNND, and HTD-Net on most data, and ACE on P1, V3, and asbestos, which further confirms the above conclusions obtained from ROC curves. For an in-depth analysis of the reasons behind this, ACE is mainly influenced by its high sensitivity to spectral distortion, and CSCR obtains unsatisfactory performance mainly limited by the amounts of target dictionaries. As for CNND and HTD-Net, they face challenges of data misalignment and limited sample diversity, respectively. The results illustrate that direct knowledge transfer across domains is infeasible, which further demonstrates the effectiveness of TLNSS combining transfer learning and diverse spectral synthesis. Moreover, we observe that all comparing methods exhibit lower robustness on Pavia than on other datasets. This is primarily attributed to the limited image quality resulting from the extremely severe spectral mixture and spatial–spectral irreconcilable pansharpening mechanism, as well as the inaccurate ground truth arising from the absence of labeling some asbestos-containing areas that are not classified as buildings.

For fairness, Table II lists AUC, $P_f^{\min}(P_d = 1)$, and average rank of all comparing methods, and the superscript numbers in statistical test represent the total rank after Nemenyi test. It can be seen that our TLNSS achieves the best performance in comparison with other detection algorithms on the testing images. Although TLNSS identifies less target pixels at an extremely low P_f as shown in Fig. 9, it usually has a higher AUC and lower $P_f^{\min}(P_d = 1)$, providing a precise supplementary evaluation for subjective ROC curves and separability maps. The AUC and $P_f^{\min}(P_d = 1)$ values obtained by the competitors further confirm their shortcomings in interested target highlighting and false alarm pixels elimination. Moreover, we find that HTD-Net obtains well detection accuracy for P1, which is barely observable in the visual map, and the same phenomenon also occurs in CSCR. The good performance but poor visualization indicate that single evaluation criterion is unreliable. For a comprehensive analysis, the p-values given by the Friedman test with $\alpha = 0.05$ indicate that the comparing detection methods are clearly different, and the ranking tests show that the overall performance of TLNSS surpasses that of other methods. The above results and analysis verify the superiority of TLNSS in target identification and background suppression, indicating the effectiveness of our proposed end-to-end detection framework based on reconstruction residuals.

2) Cross-Data Detection: Fig. 11 shows the detection maps obtained by different algorithms on Viareggio T2. Similar to the results on single-data detection, the statistical detectors and TLNSS have better performance in background suppression than CSCR, CNND, and HTD-Net. Due to the cross-data setup, the false alarm pixels on Viareggio T2 of all methods increase in comparison with that on Viareggio T1. By zooming in on region of interest, CEM, NGDS-SNR, NGR-SNR, DS-SA^2, R-SA^2, and TLNSS correctly locate five kinds of targets, and the target confidence of TLNSS seems to be much stronger. Besides, we notice that ACE could additionally detect V3 in comparison with the result on Viareggio T1. The possible reason is that the inserted background changes (refer to [49, Fig. 6(b)] for details) are in a subspace more orthogonal to the constructed detector, which could enlarge the interclass distinction and highlight what was previously unclear. This result again demonstrates the sensitivity of ACE to spectral variation.

Figs. 12 and 13 show the ROC curves and separability maps obtained by different detection algorithms on Viareggio T2, respectively. It is obvious that the ROC curves of five objects on Viareggio T2 are worse than that on Viareggio T1 but still acceptable. At a false alarm rate less than 10^{-2}, the statistical methods and TLNSS could identify more target pixels than the others in general. However, as P_f increases, the proposed TLNSS algorithm shows satisfactory detection performance, which always identifies all target pixels first or second. The separability maps further prove the aforementioned analysis, and even for P2 with the highest detection difficulty, TLNSS

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TABLE II
QUANTITATIVE COMPARISON FOR DIFFERENT ALGORITHMS ON THE EXPERIMENTAL DATASETS. THE BEST RESULT IS HIGHLIGHTED IN BOLD

Data		Accuracy	CEM	ACE	NGDS-SNR	NGR-SNR	DS-SA^2	R-SA^2	CSCR	CNND	HTD-Net	TLNSS
Viareggio T1	P1	AUC	0.9998	0.7301	0.9992	0.9992	0.9996	0.9997	0.9865	0.8044	0.9942	**0.9999**
		$P_f^{min}(P_d=1)$	7.9419e-04	0.9738	0.0031	0.0027	0.0020	0.0017	0.0657	0.9816	0.0225	**5.9860e-04**
	P2	AUC	0.9982	0.9379	0.9556	0.9793	0.9720	0.9753	0.7839	0.4555	0.8496	**0.9996**
		$P_f^{min}(P_d=1)$	0.0282	0.9759	1.0000	0.3902	0.4446	0.3915	0.4880	0.9945	0.9737	**0.0054**
	V1	AUC	0.9996	0.9422	0.9992	0.9993	0.9994	0.9995	0.9182	0.7953	0.8597	**0.9998**
		$P_f^{min}(P_d=1)$	0.0031	0.8125	0.0071	0.0052	0.0067	0.0045	0.5446	0.9929	0.4779	**5.5708e-04**
	V3	AUC	0.9999	0.7196	0.9997	0.9998	0.9999	**1.0000**	0.9529	0.7149	0.9321	**1.0000**
		$P_f^{min}(P_d=1)$	2.4298e-04	0.9591	9.0672e-04	7.1115e-04	4.0891e-04	1.5408e-04	0.0835	0.9978	0.1736	**8.2967e-05**
	V4	AUC	0.9998	**0.9999**	0.9995	0.9995	0.9996	0.9995	0.9783	0.7031	0.8643	**0.9999**
		$P_f^{min}(P_d=1)$	6.8158e-04	6.8751e-04	0.0053	0.0056	0.0040	0.0044	0.1628	0.9974	0.8766	**2.4300e-04**
San Diego		AUC	0.9964	0.9718	0.9939	0.9959	0.9925	0.9945	0.9927	0.6466	0.9613	**0.9975**
		$P_f^{min}(P_d=1)$	0.0912	1.0000	0.2481	0.1698	0.2760	0.1710	0.1434	0.7990	0.6130	**0.0091**
Pavia		AUC	0.8709	0.7679	0.8754	0.8881	0.8655	0.8795	0.8553	0.4617	0.6558	**0.9471**
		$P_f^{min}(P_d=1)$	0.9981	1.0000	1.0000	1.0000	1.0000	1.0000	**0.8215**	0.9999	1.0000	0.8313
Average		AUC	0.9807	0.8670	0.9747	0.9802	0.9755	0.9783	0.9240	0.6545	0.8738	**0.9920**
		$P_f^{min}(P_d=1)$	0.1603	0.8174	0.3235	0.2249	0.2477	0.2247	0.3299	0.9662	0.5910	**0.1210**
Statistical test	AUC		$\alpha=0.05$, p-value=1.2537e-14									
			$2.71^{1,2}$	$7.29^{2,3}$	$5.43^{2,3}$	$4.14^{1,2}$	4.86^{2}	$3.71^{1,2}$	$7.57^{2,3}$	9.86^{3}	$8.43^{2,3}$	**1.00^{1}**
	$P_f^{min}(P_d=1)$		$\alpha=0.05$, p-value=1.8445e-10									
			$2.29^{1,2}$	$8.14^{2,3}$	$7.14^{2,3}$	$5.14^{2,3}$	5.00^{2}	$4.14^{1,2}$	$5.86^{2,3}$	8.86^{3}	$7.29^{2,3}$	**1.14^{1}**

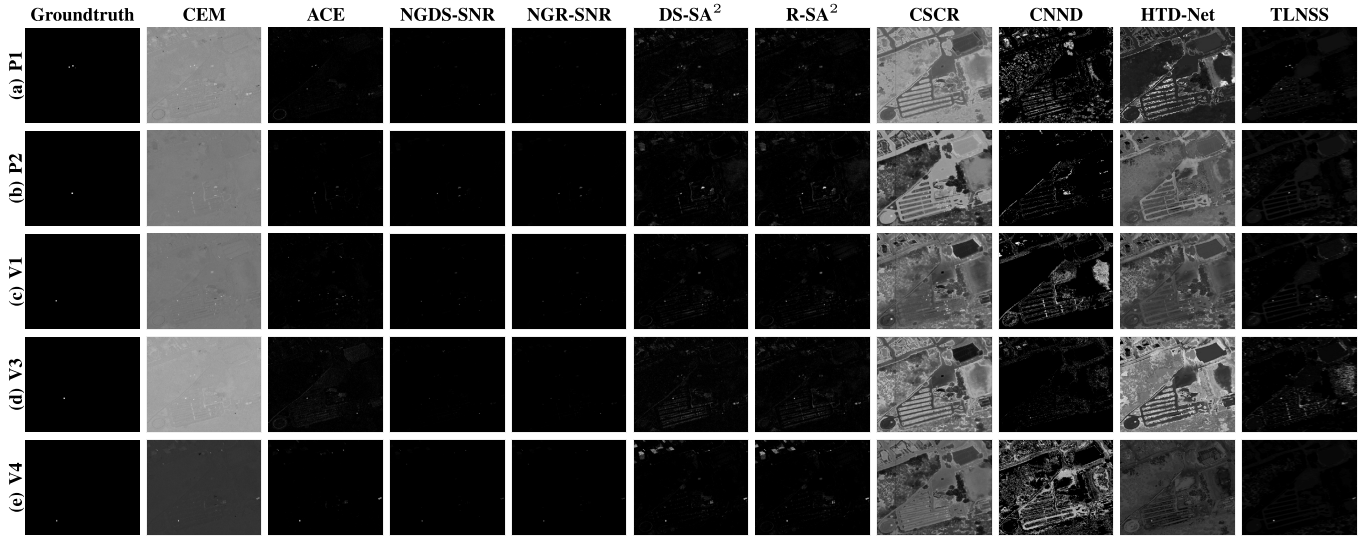


Fig. 11. Detection maps of different competitors on Viareggio T2. From left to right: ground truth, detection maps of CEM, ACE, NGDS-SNR, NGR-SNR, DS-SA^2, R-SA^2, CSCR, CNND, HTD-Net, and TLNSS. Best view via zoomed-in view. (a) P1. (b) P2. (c) V1. (d) V3. (e) V4.

could successfully distinguish it. The possible reason is that TLNSS trained with the simulated nonlinear spectra could provide more advisable spectral variants, thus better dealing with the inconsistency between testing scenarios and training samples.

Table III shows AUC, $P_f^{min}(P_d=1)$, and average rank of different detectors on Viareggio T2 for objective evaluation. The proposed TLNSS could carry out target detection with higher accuracy in most cases. In comparison with single-data detection on Viareggio T1, CEM, ACE, NGDS-SNR, NGR-SNR, DS-SA^2, R-SA^2, CSCR, CNND, HTD-Net, and TLNSS for cross-data detection on Viareggio T2 show AUC drops of 3.97%, −6.75%, 5.93%, 4.85%, 5.71%, 4.34%, 22.64%, 11.18%, 4.97%, and 1.97%, respectively, and $P_f^{min}(P_d=1)$ increases of 41.16%, −9.25%, 48.45%, 50.03%, 41.11%, 23.62%, −0.68%, 4.87%, and 23.28% in average, respectively.

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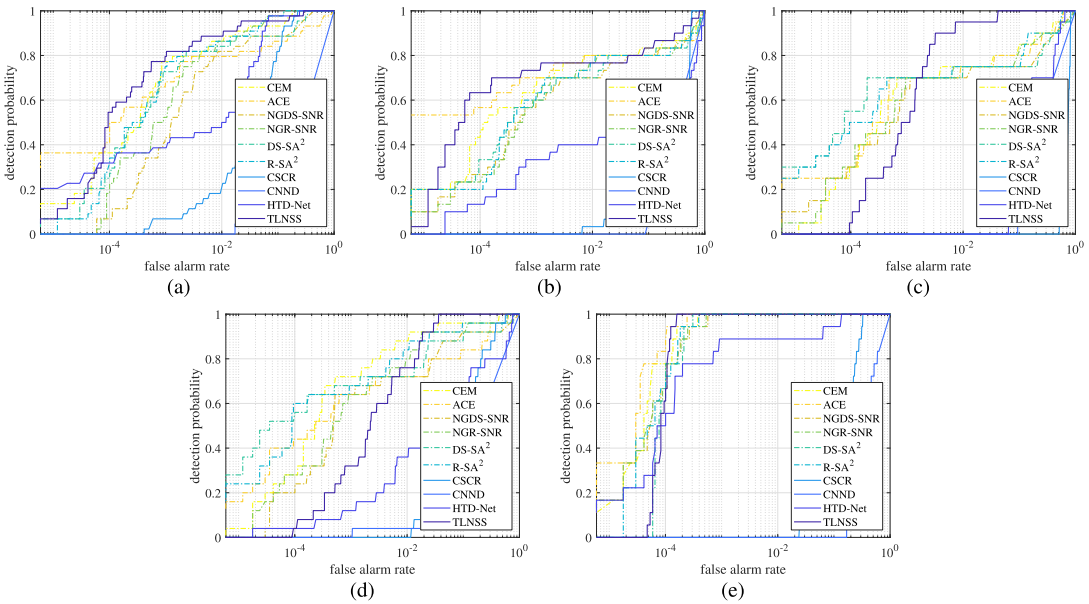


Fig. 12. ROC curves of different detection algorithms on Viareggio T2. (a) Viareggio T2 P1. (b) Viareggio T2 P2. (c) Viareggio T2 V1. (d) Viareggio T2 V3. (e) Viareggio T2 V4.

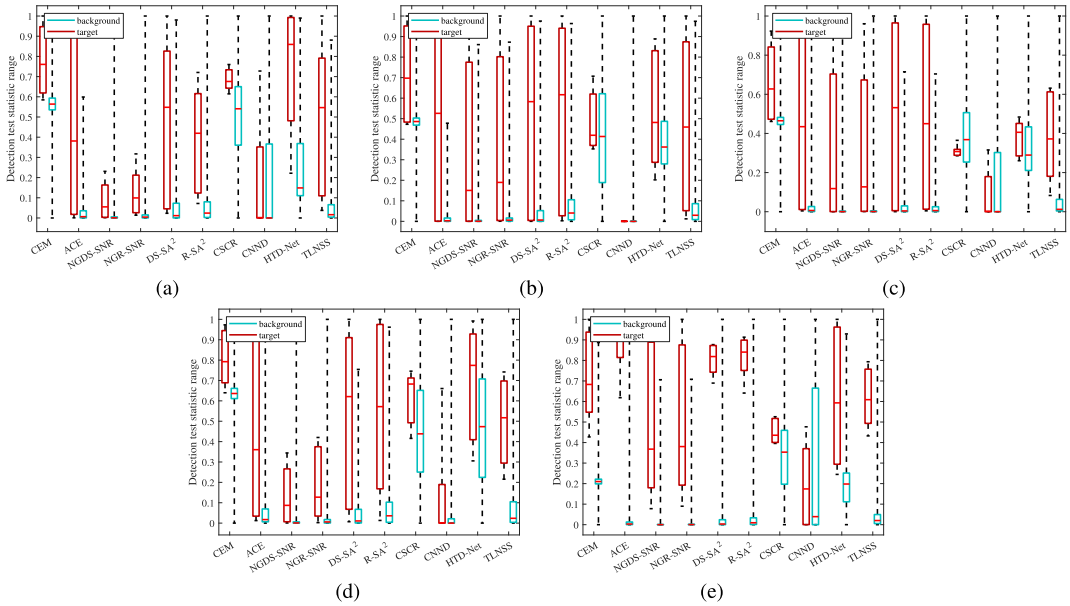


Fig. 13. Separability maps of different detection algorithms on Viareggio T2. (a) Viareggio T2 P1. (b) Viareggio T2 P2. (c) Viareggio T2 V1. (d) Viareggio T2 V3. (e) Viareggio T2 V4.

Besides two abnormal negative values, the performance degradation of TLNSS is generally weaker than that of other comparing methods, indicating its stronger reliability in cross-data detection. The opposite improvements of ACE are mainly due to the effects of insertion background on the detection space. As for CNND, the model was trained neither with Viareggio T1 nor with Viareggio T2, so there is no absolute relationship between their performances.

3) Complexity Analysis: In order to evaluate the efficiency of the proposed method, Table IV lists the computational costs of different comparing algorithms on Viareggio T1, San Diego, Pavia, and Viareggio T2. For CEM, ACE, NGDS-SNR, NGR-SNR, DS-SA^2, R-SA^2, and CSCR on Viareggio T2, the detectors are again constructed using the correlation or covariance matrix derived from Viareggio T1 for convenience. For deep learning-based CNND, HTD-Net, and TLNSS, the computational costs are categorized into training and testing phases, and the training phase of TLNSS is further classified into feature extraction subnetwork (E) training and feature fusion-spectra restoration subnetwork (R) training. CNND and TLNSS (E) are trained only once regardless of the target types. HTD-Net and TLNSS (R) are required to be retrained when the interested target changes, i.e., seven times in correspondence with seven targets in single-data detection. It can be found that the average training time of CNND is less expensive than that of TLNSS, but it cannot achieve outstanding detection performance, and HTD-Net with cumbersome preprocessing steps costs

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TABLE III

QUANTITATIVE COMPARISON FOR DIFFERENT DETECTION ALGORITHMS ON VIARREGIO T2. THE BEST RESULT IS HIGHLIGHTED IN BOLD

Data		Accuracy	CEM	ACE	NGDS-SNR	NGR-SNR	DS-SA^2	R-SA^2	CSCR	CNND	HTD-Net	TLNSS
Viareggio T2	P1	AUC	0.9876	0.9323	0.9661	0.9917	0.9702	**0.9918**	0.9426	0.6199	0.9762	0.9884
		$P_f^{min}(P_d=1)$	0.1864	1.0000	0.4381	0.1350	0.3435	**0.1276**	0.2256	0.9697	0.2425	0.2883
	P2	AUC	0.8967	0.8715	0.8780	0.8755	0.8783	0.8745	0.5709	0.4630	0.6845	**0.9223**
		$P_f^{min}(P_d=1)$	0.8543	1.0000	1.0000	1.0000	1.0000	0.9606	**0.5851**	0.9871	0.9809	0.8041
	V1	AUC	0.9346	0.9258	0.8804	0.9050	0.8871	0.9276	0.2585	0.6248	0.7739	**0.9966**
		$P_f^{min}(P_d=1)$	0.6098	0.6183	1.0000	1.0000	1.0000	0.5910	0.8253	0.9827	0.6498	**0.0420**
	V3	AUC	0.9803	0.9376	0.9325	0.9628	0.9494	0.9630	0.8707	0.6608	0.8280	**0.9932**
		$P_f^{min}(P_d=1)$	0.4399	0.6410	1.0000	0.7860	0.6154	0.7782	0.5618	0.9971	0.7576	**0.0364**
	V4	AUC	**0.9999**	**0.9999**	**0.9999**	**0.9999**	**0.9999**	**0.9999**	0.8455	0.5454	0.9889	**0.9999**
		$P_f^{min}(P_d=1)$	3.7930e-04	2.4299e-04	5.6895e-04	5.3339e-04	3.0818e-04	3.9115e-04	0.3276	0.9932	0.1371	**1.5409e-04**
Average		AUC	0.9598	0.9334	0.9314	0.9470	0.9370	0.9514	0.6976	0.5828	0.8503	**0.9801**
		$P_f^{min}(P_d=1)$	0.4182	0.6519	0.6877	0.5843	0.5918	0.4916	0.5051	0.9860	0.5536	**0.2342**
Statistical test		AUC	$\alpha=0.05$, p-value=5.6239e-11									
			$2.00^{1,2}$	$5.40^{1,2}$	5.80^2	$4.00^{1,2}$	$4.80^{1,2}$	$5.00^{1,2}$	8.80^2	9.8000^2	8.00^2	**1.40**^{1}
		$P_f^{min}(P_d=1)$	$\alpha=0.05$, p-value=0.0138									
			2.40^1	$5.20^{1,2}$	7.80^2	7.80^2	$5.40^{1,2}$	$5.40^{1,2}$	$4.80^{1,2}$	7.60^2	$6.00^{1,2}$	$2.60^{1,2}$

TABLE IV

COMPUTATIONAL COSTS (IN SECONDS) OF DIFFERENT DETECTION ALGORITHMS ON VIAREGGIO T1, SAN DIEGO, PAVIA, AND VIAREGGIO T2

Data		CEM	ACE	NGDS-SNR	NGR-SNR	DS-SA^2	R-SA^2	CSCR	CNND		HTD-Net^2		TLNSS^3		
									Train	Test	Train	Test	Train(E)	Train(R)	Test
Viareggio T1	P1	0.89	4.96	0.14	0.11	11.42	11.78	60.12	939.60	6.96	380.35	191.57	456.58	215.75	10.28
	P2	1.35	8.27	0.28	0.22	22.85	23.43	80.26	/[1]	7.16	380.44	197.14	/	180.24	19.32
	V1	1.94	9.51	0.41	0.34	36.18	35.94	103.49	/	10.67	435.23	196.90	/	230.47	13.94
	V3	2.19	10.24	0.36	0.31	35.16	36.05	99.65	/	10.68	436.28	199.92	/	134.78	11.24
	V4	1.77	11.38	0.36	0.30	56.70	77.27	94.23	/	12.16	434.07	197.21	/	105.11	16.75
San Diego		0.01	0.05	0.02	0.01	2.98	2.98	2.05	/	15.69	328.71	17.73	/	42.13	2.27
Pavia		0.40	5.17	1.55	1.35	369.94	383.47	368.70	/	20.44	371.25	298.90	/	244.25	20.53
Viareggio T2	P1	0.73	5.13	0.12	0.11	11.52	11.52	73.24	/	8.55	/	194.86	/	/	9.27
	P2	1.48	7.34	0.24	0.21	23.35	23.78	89.35	/	6.98	/	192.54	/	/	11.39
	V1	1.89	10.10	0.35	0.32	35.03	35.42	105.49	/	10.45	/	199.30	/	/	17.28
	V3	2.03	9.72	0.36	0.31	35.22	35.73	110.32	/	10.69	/	197.30	/	/	15.76
	V4	1.66	9.39	0.36	0.30	34.43	35.56	98.68	/	12.18	/	199.79	/	/	16.31

[1] / means that no corresponding procedure is required.
[2] The training time of HTD-Net does not contain data preparation, which takes at least 300 seconds in the experimental setup.
[3] The computation cost of training TLNSS (R) includes that spends on nonlinear spectral synthesis.

much more time than TLNSS. Moreover, regardless of network training, CEM, ACE, NGDS-SNR, NGR-SNR, CNND, and TLNSS have higher efficiency than DS-SA^2, R-SA^2, CSCR, and HTD-Net. The possible reasons include: the computational complexities of DS-SA^2 and R-SA^2 are highly related to the spatial size of the testing hyperspectral images; CSCR needs to solve complex optimization problems with ℓ_1-norm and ℓ_2-norm regularizations; and HTD-Net constructs multiple positive and negative pixel-pairs for each testing pixel, increasing the number of testing samples several times compared to TLNSS. As a consequence, HTD-Net bears a significantly heavier computational burden than TLNSS during the testing phase. Although CEM, ACE, NGDS-SNR, NGR-SNR, and CNND have advantages in computational costs, their performances are restrained in handling spectral distortion. In conclusion, our proposed TLNSS could obtain outstanding detection accuracy with decent efficiency, especially in target identification across data.

C. Ablation Analysis

The proposed TLNSS method comprises a nonlinear spectral synthesizer and an asymmetrical autoencoder, and this

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TABLE V
IMPACTS OF SPECTRAL MIXTURE MODEL ON TARGET DETECTION

Data	Accuracy	LMM	BMM
Viareggio T1 P1	AUC	0.9998	0.9999
	$P_f^{min}(P_d = 1)$	0.0012	5.9860e-04
Viareggio T1 P2	AUC	0.9994	0.9996
	$P_f^{min}(P_d = 1)$	0.0074	0.0054
Viareggio T1 V1	AUC	0.9988	0.9998
	$P_f^{min}(P_d = 1)$	0.0013	5.5708e-04
Viareggio T1 V3	AUC	0.9991	1.0000
	$P_f^{min}(P_d = 1)$	0.0076	8.2967e-05
Viareggio T1 V4	AUC	0.9999	0.9999
	$P_f^{min}(P_d = 1)$	0.0018	2.4300e-04
San Diego	AUC	0.9933	0.9975
	$P_f^{min}(P_d = 1)$	0.2596	0.0091
Pavia	AUC	0.9266	0.9471
	$P_f^{min}(P_d = 1)$	0.8468	0.8313

TABLE VI
IMPACTS OF ATTBLOCK ON TARGET DETECTION

Data	Accuracy	Mul	MulAtt	TLNSS
Viareggio T1 P1	AUC	0.9989	0.9996	0.9999
	$P_f^{min}(P_d = 1)$	0.0096	0.0026	5.9860e-04
Viareggio T1 P2	AUC	0.8846	0.9865	0.9996
	$P_f^{min}(P_d = 1)$	0.4862	0.1649	0.0054
Viareggio T1 V1	AUC	0.9998	0.9997	0.9998
	$P_f^{min}(P_d = 1)$	0.0024	0.0023	5.5708e-04
Viareggio T1 V3	AUC	0.9734	0.9989	1.0000
	$P_f^{min}(P_d = 1)$	0.2383	0.0077	8.2967e-05
Viareggio T1 V4	AUC	0.9978	0.9977	0.9999
	$P_f^{min}(P_d = 1)$	0.0447	0.0527	2.4300e-04
San Diego	AUC	0.9948	0.9951	0.9975
	$P_f^{min}(P_d = 1)$	0.0284	0.0247	0.0091
Pavia	AUC	0.9287	0.9444	0.9471
	$P_f^{min}(P_d = 1)$	0.8714	0.8568	0.8313

TABLE VII
IMPACTS OF DIFFERENT FEATURE COMBINATION ON TARGET DETECTION

Data	Accuracy	FF^3	FF^{13}	FF^{23}	TLNSS
Viareggio T1 P1	AUC	0.9984	0.9997	0.9998	0.9999
	$P_f^{min}(P_d = 1)$	0.0113	0.0024	0.0014	5.9860e-04
Viareggio T1 P2	AUC	0.7806	0.8796	0.8650	0.9996
	$P_f^{min}(P_d = 1)$	0.3277	0.3058	0.3012	0.0054
Viareggio T1 V1	AUC	0.9634	0.9997	0.9881	0.9998
	$P_f^{min}(P_d = 1)$	0.1754	0.0022	0.1059	5.5708e-04
Viareggio T1 V3	AUC	0.5344	0.6113	0.5734	1.0000
	$P_f^{min}(P_d = 1)$	0.5121	0.4786	0.5030	8.2967e-05
Viareggio T1 V4	AUC	0.7759	0.8462	0.9769	0.9999
	$P_f^{min}(P_d = 1)$	0.4162	0.4178	0.2988	2.4300e-04
San Diego	AUC	0.9917	0.9964	0.9965	0.9975
	$P_f^{min}(P_d = 1)$	0.0397	0.0264	0.0205	0.0091
Pavia	AUC	0.8461	0.8524	0.8568	0.9471
	$P_f^{min}(P_d = 1)$	0.9794	0.8374	0.9322	0.8313

section performs an ablation study on both parts to evaluate the effectiveness and contributions of our work.

1) Impacts of Spectral Mixture Model: To experimentally illustrate the effectiveness of BMM on HTD, we additionally exploit LMM to generate mixing spectra for subsequent network optimization, and the quantitative results are given in Table V. It can be observed that BMM performs better in overall detection accuracy and achieves a significantly smaller false alarm rate at $P_d = 1$. This finding validates that the nonlinear spectra simulated by BMM could more effectively describe real-world scenarios and model spectral distortion as comprehensively as possible, thus helping to identify interested targets from complex backgrounds.

2) Impacts of Asymmetrical Autoencoder: In this section, we analyze the effectiveness of feature fusion subnetwork on HTD and display the numerical results in Table VI. The baseline model (denoted as Mul) is built up by discarding all AttBlocks, and the transition model (denoted as MulAtt) removes the residual structure of AttBlock. Embedding features by straightforward fusion may contain redundant interference, which brings unnecessary discrepancy to feature characterization. MulAtt integrates multilayer representations in a complementary way, but some important features missed by the attention map will not be transmitted to the restoration phase, nor can it be modified. Our proposed Attblock with residual learning consists of two branches responsible for trunk feature transfer and informative feature highlighting, which could resist information loss and achieve more accurate restoration. Moreover, Fig. 14 visualizes the feature maps obtained by three blocks of Mul, MulAtt, and TLNSS, and each feature is sent forward to the following convolution layer to obtain the same dimension for intuitive visualization comparison. It can be observed that the distinctions between the testing samples and target prior of Mul, MulAtt, and TLNSS mainly lie on all features with the same importance, partial crucial features, and all features with informative keypoints particularly emphasized, respectively. These results illustrate that AttBlock could aid in discriminative feature extraction.

Furthermore, we investigate the effects of feature fusion from different ConvBlocks on HTD, i.e., integrating output features of the ith and jth ConvBlock in the feature fusion subnetwork (denoted as FF^{ij}). The detailed results are reported in Table VII. It can be seen that the multilayer feature fusion could always provide more effective information than a single high-level representation, indicating the significance of low- and high-level feature complementarity for reverse feature reconstruction.

Finally, considering that the final detection results are obtained by integrating three residual maps, we carry out experiments to analyze the contribution of each map.

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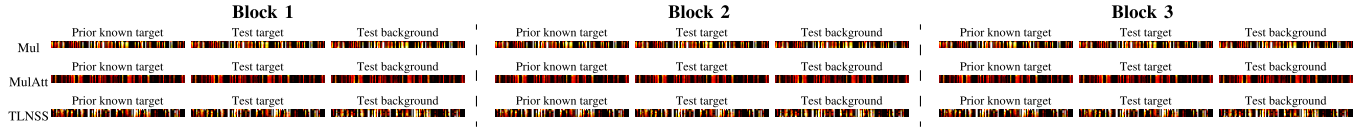


Fig. 14. Feature maps obtained by three blocks of Mul, MulAtt, and TLNSS. Best view via zoomed-in view.

TABLE VIII
IMPACTS OF DIFFERENT RESIDUAL MAPS ON TARGET DETECTION

Data	Accuracy	L_{rec}^1	L_{rec}^2	L_{rec}^3	TLNSS
Viareggio T1 P1	AUC	0.9999	0.9998	0.9997	0.9999
	$P_f^{min}(P_d=1)$	2.6670e-04	6.7263e-04	0.0018	5.9860e-04
Viareggio T1 P2	AUC	0.9994	0.9983	0.9907	0.9996
	$P_f^{min}(P_d=1)$	0.0049	0.0236	0.1167	0.0054
Viareggio T1 V1	AUC	0.9999	0.9999	0.9997	0.9998
	$P_f^{min}(P_d=1)$	2.3705e-04	2.6669e-04	0.0022	5.5708e-04
Viareggio T1 V3	AUC	0.9999	0.9998	0.9933	1.0000
	$P_f^{min}(P_d=1)$	2.9631e-05	7.7041e-04	0.0370	8.2967e-05
Viareggio T1 V4	AUC	0.9998	0.9975	0.9970	0.9999
	$P_f^{min}(P_d=1)$	1.0076e-04	0.0089	0.0453	2.4300e-04
San Diego	AUC	0.9976	0.9975	0.9972	0.9975
	$P_f^{min}(P_d=1)$	0.0092	0.0085	0.0103	0.0091
Pavia	AUC	0.9358	0.9261	0.8976	0.9471
	$P_f^{min}(P_d=1)$	0.8154	0.8951	0.8972	0.8313

Specifically, the residual map between the ith ConvBlock and the ith DconvBlock is denoted as L_{rec}^i, and Table VIII lists the quantitative accuracy. It can be observed that L_{rec}^1 generally provides more reliable detection results than L_{rec}^2 and L_{rec}^3, probably because low-level features with more spectral bands possess stronger discriminative capacity. Moreover, we find that residual map fusion would bring slight AUC improvement and $P_f^{\min}(P_d = 1)$ degradation. According to existing analysis, we tentatively conclude that there is no necessity to apply very deep features with little spectral knowledge for the final detection from the perspectives of efficiency and accuracy.

IV. CONCLUSION

Oriented to deep-learning-based HTD tasks, this article presents an end-to-end framework based on TLNSS to address the issue of training samples with limited quantity and diversity. To simulate spectral vectors modeling spectral distortion in real scenarios as closely as possible, this article proposes a nonlinear spectral synthesis method to describe complicated interactions between materials. Then, a novel asymmetric autoencoder paradigm is constructed for HTD, which applies multiscale feature extraction and simplified spectra restoration architecture for efficient and effective network optimization. It holistically transforms target detection to spectra restoration and controls the reconstruction residuals of target and background by a specifically designed multilevel loss. Moreover, we complementarily integrate features from different layers to achieve more accurate restoration as expected. Experimental results and analysis of single-data and cross-data detection indicate the superiority of TLNSS, and further ablation studies validate the effectiveness of each particularly designed block.

In future research, we intend to investigate effective and efficient 3-D data simulation methods considering both spectral reflectance and spatial structure, which can help to better handle spectral distortions and further promote the advancement of HTD.

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